guardrails.py - Hard Safety Rules

What this file does (big picture)

guardrails.py is the agent's non-negotiable policy layer.

Business rules encoded here:

1) Do not diagnose medical conditions
2) Do not recommend changing prescriptions
3) Escalate emergencies
4) Do not promise refunds
5) Do not expose another customer's order information

Two checkpoints: check_input (before we answer) and check_output (after a draft exists - including future AI drafts).

Professional habit: put safety in its own module so every channel (CLI today, Twilio later) can call the same rules.

New words you will see in this file

guardrail	A forced safety check that overrides clever or unsafe replies.
allow / block / escalate	The three actions: proceed, refuse safely, or hand to a human.
input check	Inspect the customer's message before classification.
output check	Inspect our draft reply before the customer sees it.
regex	A text pattern (re.search) used to catch refund-promise wording.
@property	A method used like a field - result.allowed reads as a True/False flag.

Flow while the program runs

check_input(question) -> allow, block, or escalate.

If allow: build draft elsewhere -> check_output(draft) / apply_output_guardrails.

Blocked drafts are replaced with a safe message; emergencies escalate first.

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1-17

```
"""Guardrails for the sleep-company AI agent..."""
```

Plain English:
Module docstring lists the five business rules and explains input vs output checks.

19-23

```
imports: re, dataclass, Optional
```

Plain English:
Regex for pattern matching, dataclass for the result object, Optional for missing reasons.

25-28

```
ALLOW / BLOCK / ESCALATE = "allow" / "block" / "escalate"
```

Plain English:
Action constants so main.py compares names, not magic strings scattered everywhere.

30-36

```
REASON_* constants
```

Plain English:
Machine-readable why-codes for logs and future EscalationPacket notes.

39-56

```
@dataclass class GuardrailResult
```

Plain English:
Traffic-light result: action, optional reason, customer-facing message. .allowed is True only for allow.

59-61

```
def _contains_any(...)
```

Plain English:
Private helper: does the text contain any clue phrase?

66-77

```
_EMERGENCY_CLUES list
```

Plain English:
Phrases that mean stop and escalate (chest pain, suicidal language, call 911, etc.).

79-102

```
_DIAGNOSIS_CLUES and _PRESCRIPTION_CLUES
```

Plain English:
Phrases that mean refuse: diagnosing conditions or advising medication changes.

104-121

```
_OTHER_CUSTOMER_CLUES
```

Plain English:
Privacy: looking up someone else's order (husband/wife/friend) is blocked.

124-180

```
def check_input(question)
```

Plain English:
Order matters: emergency first, then prescription, diagnosis, other-customer privacy. Otherwise allow.

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185-208

```
refund / diagnosis / prescription output patterns
```

Plain English:
Patterns and clues that mean a *draft reply* broke policy (especially refund promises).

211-273

```
def check_output(draft, question)
```

Plain English:
Scan the draft for refund promises, diagnosis/prescription wording, and unexpected emails not present in the question.

276-281

```
def apply_output_guardrails(...)
```

Plain English:
Convenience wrapper: return the draft if safe, else the safe refusal message.

284-293

```
def policy_summary()
```

Plain English:
Printable list of active rules for operators and documentation.